Virtualisation · Pharmaceutical Research

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Virtualization project in an Institute pharmaceutical research In a Research Institute in the pharmaceutical field, the Virtualization is a particularly suitable solution for support heterogeneous work environments, in which applications of mathematics, biology, pharmacy, science coexist social and other disciplines with very different technical requirements. The Virtualization allows you to consolidate resources, isolate applications incompatible with each other, simplify administration and improve availability of services. Among the alternatives, they stand out VMware ESXi and Microsoft Hyper -V, two platforms with approaches different, but with solid capabilities for environments corporate.

1. Main features of VMware ESXi and most important services associated with vCenter VMware ESXi is a type hypervisor that is installed directly on the physical hardware of the server and acts as a layer of main virtualization. Its purpose is to run multiple machines isolated virtual networks on the same physical infrastructure, with a efficient CPU, memory, network and storage management.

This architecture is especially useful in organizations with great diversity of software, since it allows environments to be encapsulated different without depending on an identical configuration in all physical equipment. It should be noted that vCenter Server is not part of the hypervisor in the strict sense, but constitutes the management platform centralization of the vSphere ecosystem. From vCenter it is possible manage ESXi hosts, virtual machines, clusters, resources and operational policies from a unified console, thereby reducing very notable administrative complexity in environments medium and large. Among the most important additional services that vCenter enables, The first one that should be highlighted is vMotion, a technology that allows migrate powered-on virtual machines between hosts without interruption noticeable from the service. In a research institute, this

capacity is critical because it facilitates maintenance tasks, redistribution of loads or replacement of hardware without downtime scientific applications in production. Another essential service is Storage vMotion, which allows you to move warm up the files and disks of a virtual machine from a datastore to another while still running. This function provides great flexibility in storage management, whether for reorganize cabins, free up space or improve the performance of certain services.

vSphere HA and DRS are also essential. vSphere HA Provides high availability by restarting virtual machines on other hosts in the cluster if the physical server fails. By DRS, on the other hand, distributes the workload among hosts and optimizes the cluster resource usage, balancing CPU consumption and memory. In an environment where laboratory applications coexist, statistical analysis and document processing, these mechanisms improve both service continuity and efficiency operational.

2. Main benefits of Hyper -V Hyper-V is Microsoft's virtualization technology based on hypervisor. According to the official documentation, it allows you to run multiple isolated operating systems on the same platform physical and is aimed at both business scenarios and testing environments, consolidation and continuity of service.

One of its main benefits is the optimization of resources and cost reduction, since it allows several services to be consolidated in fewer physical servers, improving the use of hardware and reducing energy consumption, space and costs of operation and maintenance. This advantage has a direct impact on centers with multiple departments and software needs scattered.

Another relevant benefit is its extensive system support guest operatives. Hyper -V supports multiple versions of Windows and also Linux and FreeBSD distributions, which makes it makes it a versatile solution for organizations where There is homogeneity of client or server platforms. This

capacity is especially interesting in scientific environments, where some programs require Windows and others run better on Linux. Added to this is high availability and operational continuity, since that Hyper -V in Windows Server incorporates mechanisms such as live migration, migratable storage and recovery options in the face of disasters. This allows intervention on the infrastructure without necessarily stop workloads, a feature very valuable when handling critical applications for research or data management. Its integration with the Microsoft ecosystem should also be noted, especially with Windows Server, PowerShell and tools centralized administration. This integration favors the automation, standardization of deployments and management easier in organizations that already work on technology Microsoft.

3. Application virtualization with Windows computers and Linux: is it a problem? The existence of computers with different operating systems does not constitutes, in itself, an impediment to the virtualization of applications. The key is in the model adopted. If the application runs centrally on servers and the user accesses remotely to it, the diversity of the client operating system loses much of its importance, since what is relevant is the compatibility between the server and the application, not so much between the application and each individual terminal.

However, if you opt for application virtualization dependent on the local operating system, then they do appear relevant restrictions. A clear example is Microsoft App-V, whose Supported configurations focus on Windows environments and Windows Server. Therefore, from a technical point of view, the Windows/Linux heterogeneity does not make the project impossible, but it does significantly conditions the chosen technology.

4. Impact of the high consumption of graphic resources and reports The fact that the Institute's applications generate graphics complex visualizations or intensive reports does affect the virtualization, because this type of load requires careful attention special to hardware sizing, especially in CPU, memory, storage, and in some cases GPU. Hyper-V explicitly documents partition-related functions GPU and device mapping, demonstrating that loads Intensive graphics require specific mechanisms and should not be be treated as if they were simple office applications. Therefore, the virtualization of these applications is viable, but not trivial. If the infrastructure design does not contemplate acceleration adequate and correct resource planning, performance can degrade significantly. In a research institute This is especially sensitive, because a drop in performance in data processing or reporting can compromising both productivity and operational reliability.

5. Remote Desktop Services: what they consist of and main limitation Remote Desktop Services (RDS) is a platform for Windows Server intended to offer desktops and applications centralized and secure way. In this model, applications are run on servers in the data center and the user accesses their interface via remote connection, making it easy to manage, patching, security and access control from multiple locations. Its main limitation, in the context presented, is that it does not eliminate the intrinsic incompatibilities of certain applications or special hardware needs. That is, RDS centralizes and simplifies software delivery, but does not convert automatically makes an application suitable for remote access requires intense graphics acceleration, very specific peripherals or inappropriate behavior in multi-user environments. This conclusion is inferred from the centralization model described

by Microsoft and the need to consider different profiles depending on compatibility and performance.

6. VMware solution to encapsulate traffic TCP/IP storage If the organization decides to use VMware and wants to encapsulate the storage traffic over TCP/IP instead of using a specialized fiber channel, the appropriate solution is iSCSI. The Broadcom documentation indicates that ESXi can connect to external SAN storage using the iSCSI protocol, precisely designed to transport SCSI commands over IP networks.

In this area, ESXi provides three main modes of connection: iSCSI software, where protocol processing is performed by software on the host itself; iSCSI hardware dependent, in which a hardware adapter intervenes but with a dependency from the VMware networking stack; and independent iSCSI hardware, based in more autonomous and specialized adapters. This classification allows you to choose between greater flexibility, lower cost or greater processing download according to project needs.

7. Types of virtual controllers in Hyper -V In Hyper-V, one of the most important distinctions is the existing between synthetic devices and inherited or emulated ones. The Synthetic devices are specific to Hyper-V and communicate via VMBus, which reduces overhead and improves performance compared to traditional emulation. Therefore, they are the preferred option in modern environments. Your main condition is that the guest system must have support suitable for this type of devices.

In contrast, legacy or emulated devices are maintained for reasons of compatibility with operating systems old or special settings. In the case of adapters legacy networks, Microsoft presents them as an option compatibility for systems or applications that do not support synthetic adapters. Its main condition is, precisely, that offer lower performance and are designed for

compatibility, not for optimal operation. Furthermore, the Hyper -V documentation links them to scenarios where still old generations of virtual machines are involved.

8. Safeguard mechanisms in VMware against problematic updates To prevent a faulty update from causing damage irreversible, a very reasonable measure at VMware is the creation preview of a snapshot of the virtual machine. Snapshots allow maintain a snapshot of the VM's status and facilitate quick rollback if an installation or patch causes instability or functional loss.

However, it is important to note that Broadcom warns expressly that snapshots should not be considered a method Robust long-term backup and recovery. By Therefore, the correct practice would be to combine snapshots prior to delicate changes with a real independent backup strategy.

Put directly: the snapshot serves as an airbag, but not replaces the seat belt.

9. Recommended procedures to minimize the CPU usage in Hyper -V If in tests with Hyper -V a notable drop in the processor performance under overload situations, Microsoft recommends several measures to reduce unnecessary activity in background. Among them are the updated installation of integration services, the use of synthetic adapters in instead of emulated ones, removing unused devices and review of scheduled tasks or unnecessary services within the virtual machines.

Likewise, the official guide recommends closing tools such as Server Manager or Hyper -V Manager when they are not necessary, since that can generate additional activity; disable certain services in the background on client VMs, such as Windows Search or tasks non-essential scheduled; and maintain prudent settings of the number of virtual processors to avoid

overprovisioning. All of this aims to reduce the consumption of CPU that brings no real value to the workload.

Finally, Microsoft also advises properly configuring antivirus exclusions on Hyper -V hosts and avoid running roles extras in the root partition if you are looking for the best performance possible. In virtualization environments, host cleanup and reduction of accessory processes are basic measures to avoid that the CPU is wasted on tasks unrelated to the computer's own function.

hypervisor.

Conclusion Q It can be said that both VMware ESXi and Hyper -V are mature and valid solutions for a virtualization project in a pharmaceutical research institute. VMware stands out especially for the strength of its vSphere ecosystem and the advanced centralized management, high availability and hot migration enabling vCenter. Hyper-V, for its part, offers a very competitive proposal in Microsoft environments, with good integration, support for multiple guest systems and adequate tools for operational continuity and optimization of resources.

Regarding application virtualization, the coexistence of Windows and Linux does not prevent the project, but forces you to select carefully technology and delivery architecture. Furthermore, the loads with high graphic demand do strongly condition the infrastructure design. Therefore, the final decision should not based only on the name of the platform, but on the appropriateness between technology, types of applications, level of criticality of the service and real profile of the Institute's users.